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Burials and engravings in a small-brained hominin, *Homo naledi*, from the late Pleistocene: contexts and evolutionary implications

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Abstract

Data from recent explorations in the Dinaledi subsystem illustrates one of the earliest examples of a mortuary practice in hominins and offers the earliest evidence of multiple interments and funerary actions, as well as evidence of the early creation of meaning making by a hominin. The hominin undertaking these behaviors was the small-brained *Homo naledi*. These data call into question several key assumptions about behavioral and cognitive evolution in Pleistocene hominins. The evidence from Dinaledi push back the temporal origins of mortuary and funerary behaviors and associate the creation of meaning making with a small-brained species and thus challenge key assumptions about the role and importance of encephalization in human evolution. This suggests that the hominin socio-cognitive niche and its relation to meaning-making activities is more diverse than previously thought. The association of these activities in subterranean spaces accessed and modified by the small brained species *Homo naledi* impacts assertions that technological and cognitive advances in human evolution are associated solely with the evolution of larger brains.

One-Sentence Summary

Burials and related meaning making in a small-brained hominin alter our understandings of human evolution.

eLife assessment

The discovery of *Homo naledi*-associated evidence for intentional burial and engravings would undoubtedly have **important** implications for our understanding of the evolution of complex cognition and behavior. Based on claims made in two related preprints by Berger et al., this study discusses the potential implications of the purported mortuary practice and symbolic behaviors claimed to be associated with the small-brained *H. naledi* remains. Unfortunately, the evidence presented in the two related submissions that the current paper entirely relies on is **incomplete** at this stage.

Main Text

Scholars have long argued that there is a qualitative difference between the abilities of *Homo sapiens* versus other Pleistocene hominins and that this difference has to do with overall brain size and neurobiological complexity. Complex behavior and the construction and use of “meaning-laden” material, e.g. burial, fire as light, engravings, ornamentation, ochre use, etc. have all been suggested to be signals of “modern” human cognitive capacity tied to a brain well above 1000cc (Kissel and Fuentes 2021; Galway-Witham, Cole, and Stringer 2019). The recent finds from the Dinaledi chamber, Rising Star Cave, South Africa indicate that large-brain-only model for complex hominin behavior no longer holds (Berger et al. 2023a,b).

It is true that overall brain size and Encephalization Quotient (EQ) increased in many populations of the genus *Homo* over the past 2 million years, and that this is generally associated with increased appearance of complex behavior. However, it is now apparent that many of these behaviors are found in places used by, or associated with, hominins that are not traditionally considered to be *Homo sapiens*. For example, engravings are found with non-*H. sapiens* hominins (Joordens et al. 2014; Mania and Mania 1988; Sirakov et al. 2010). Ochre use is documented in samples that predate *H. sapiens* (Ronen et al. 1998; Watts, Chazan, and Wilkins 2016; Dapschauskas et al. 2022) and both fire use and simple mortuary behavior are also found in association with hominins that pre-date *H. sapiens* (Carbonell and Mosquera 2006; MacDonald et al. 2021). However, one might argue that all of these instances are associated with large-brain hominins or could be the result of taphonomic issues or dating issues (Püschel et al. 2021). But this is not accurate. Substantive evidence suggests that approximately 250-350,000 years ago the later Pleistocene hominin, *Homo naledi*, a small-brained hominin, carried deceased conspecifics into difficult to access locations in the Rising Star Cave system and interred them (Berger et al. 2023 a,b) and likely produced engravings on cave walls near those areas of interment. Such actions require considerable social collaboration, coordination, and planning. And in the context of the subterranean Dinaledi system also require use of a light source; fire (Berger 2022). *H. naledi* carried out these behaviors with a brain size of less than ~600cc.

Fire use, mortuary behavior, and the evidence of engravings attributed to *H. naledi* falsify the hypothesis that only a large-brained hominin was capable of cognitively complex cultural, possibly symbolic, behavior. Recent discoveries demonstrate that at least a few populations/taxa in the genus were characterized by smaller overall brain sizes (~5-600cc) well into the later Pleistocene. Material evidence associated with these smaller brained populations overlaps with the technology and fire use (*H. floresiensis*, (Moore et al. 2009)),

mortuary behavior and the production of engravings (*H. naledi*, Berger et al. 2023a,b) exhibited by their larger brained congenics (see Fig 1). While these populations were characterized by smaller bodies than temporally sympatric hominins, their brains are not simply allometrically-scaled reductions of the larger forms (Holloway et al. 2018). This suggests that neurobiological organization rather than overall brain size, may have been one part of an early key transition within hominin evolution (8). It may be possible that the apparent gap in distribution of *Homo*/hominin fossils with an endocranial volume under 800cc between 1.0 and .3 mya in Figure 2 may be filled in with future discoveries. The open question introduced by the current evidence for *H. naledi*, and *H. floresiensis*, behavior is whether there might be one or more lineages of small brained hominins that flourished alongside larger brained forms and shared at least some aspects of the distinctive socio-cognitive hominin niche of the later Pleistocene (Galway-Witham, Cole, and Stringer 2019; Kissel and Fuentes 2021).

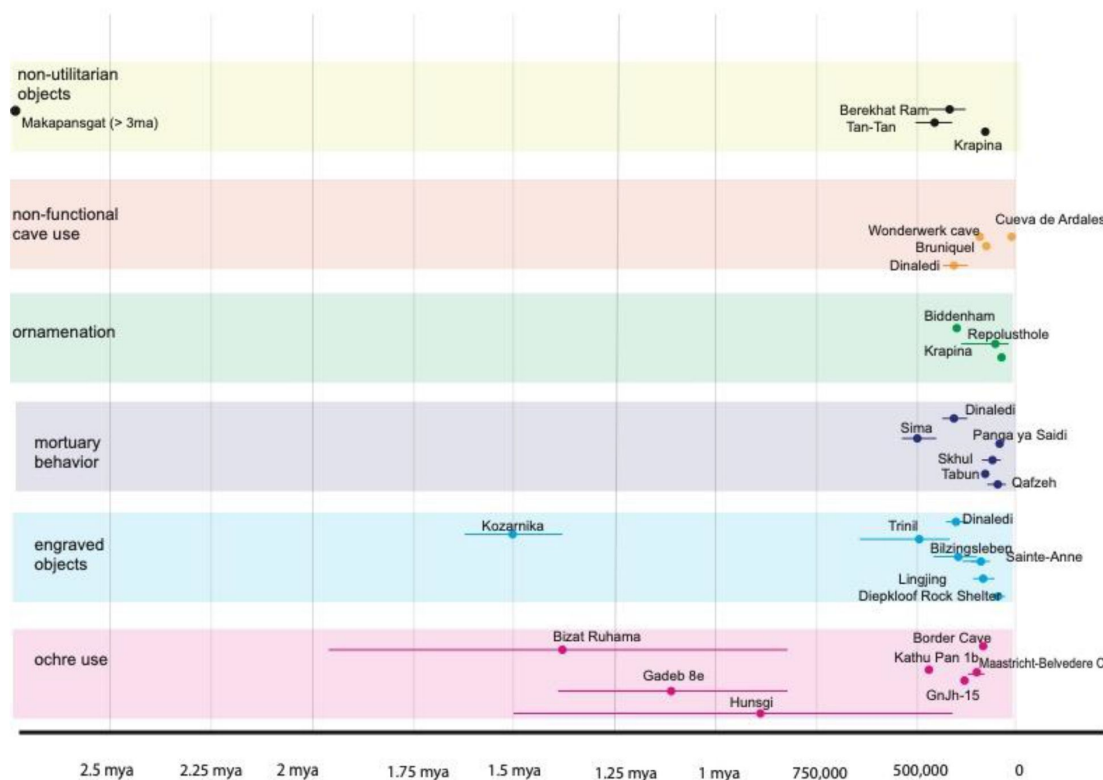


Fig. 1.

Archaeological evidence of culturally-mediated, meaning-making, behaviors.

Dots represent different sites and the error bars are the maximum and minimum dates when available. See Table for details and See supplemental material for references.

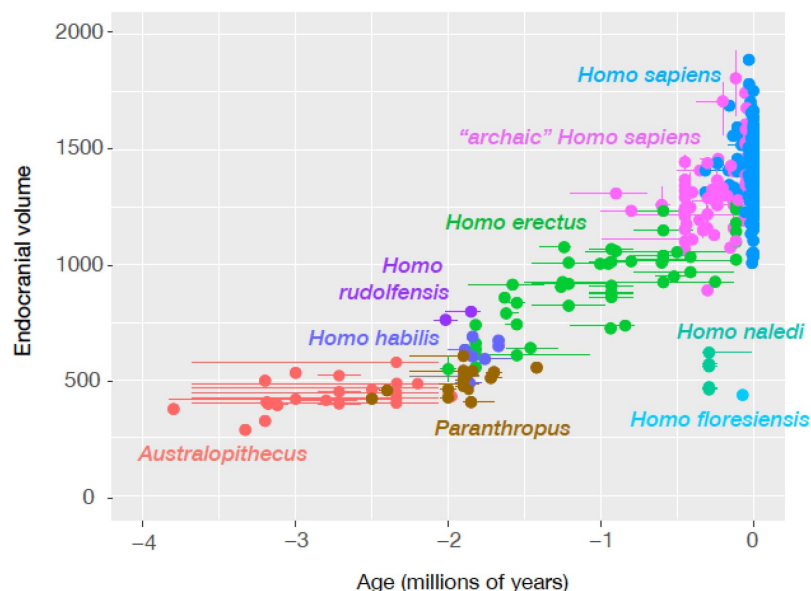


Fig. 2.

Endocranial volume estimates for hominin cranium.

Error bars represent the maximum and minimum ages for specimens when available. See supplemental material for references. Hawks, John. 2023. Endocranial volumes for fossil hominins (dataset). Figshare <https://doi.org/10.6084/m9.figshare.22743980>

Here we offer an analysis of the recent data reported for complex behavior in the small-brained *Homo naledi* and suggest a suite of implications this has for our understanding of the relationships between brain size, cognition, complex behavior and the evolution of the genus *Homo* across the Pleistocene. These implications also query the driving forces behind encephalization and its relationship to the emergence of complex behaviors in hominins and other animals.

H. naledi buried their dead

Recently Pettitt (2022) laid out the three key criteria for assessing whether or not the *Homo naledi* remains in the Dinaledi subsystem represent actual funerary behavior: a) is there an as-yet unmapped entrance into the Dinaledi Chamber? b) Is there any evidence of artificial lighting in the cave system, and c) Is there evidence that it was dead bodies, rather than body parts that were carried into the chamber? The first query has been repeatedly addressed and no other options for alternative ingress have been found or hypothesized (Elliott et al. 2021). And now Berger et al. (2023a,b) answers both (b) and (c) clearly: there is evidence of fire use in the Dinaledi system (Bower, 2022) and entire bodies rather than body parts make up much of the buried remains. Therefore, one must conclude that the evidence meets Pettitt's (2022) criterion set forth and that *H. naledi* remains in the Dinaledi subsystem are one of the two earliest examples of a mortuary practice and offer the earliest evidence of multiple interments and funerary actions in a hominin.

The locations, contexts, and the inferred behavior associated with the *H. naledi* burials, including the appearance of engravings near the interment sites, likely also demonstrate meaning-making activity (Kissel and Fuentes 2017). Mortuary and funerary practices had previously only been attributed to *Homo sapiens* but more recently are associated with a range of large-brained hominin taxa in later the Pleistocene (Figure 1). Evidence of funerary behavior is assumed to require human-like cognitive capability (Pettitt 2018). Such behavior found in a small-brained hominin suggests that increases in brain size/EQ may not be a necessary precursor for the appearance of meaning-making behavior in the hominins.

The newly described *Homo naledi* data, and the clear assignation of funerary behavior to the context, and the presence of engravings in the same space, calls into question several key assumptions about behavioral evolution in Pleistocene hominins, and the importance of

brain size evolution in general. The data emerging from the Dinaledi subsystem support an emerging argument that individual cognitive ability related to increased encephalization may not be the primary explanation of certain complex hominin behavior, such as burying of the dead, strategic use of fire for illumination, and the creation of engravings. Rather than wholly relying on increased encephalization, we suggest that a distinctive cultural, empathetic, collaborative niche dependent on increasingly complex and robust relationships between individuals has also been a primary driver in the development of key aspects of human, or human-like, behavior (Galway-Witham, Cole, and Stringer 2019; Kissel and Fuentes 2021; [McBrearty and Brooks 2000](#); [Fuentes 2017](#); [Spikins 2022](#); [DeCasien, Barton, and Higham 2022](#))

The varied treatments of the dead described in Berger et al. (2023a) is certainly indicative of *mortuary behavior*, defined as specific actions relating to death and to the treatment of the dead, and the presence of dug burials and associated engravings is likely indicative of *funerary behavior*, defined as specific activities relating to the disposal of the dead and to their subsequent commemoration ([Pettitt 2018](#)). The data reflect a scenario where members of the *H. naledi* community carried the bodies of their dead more than 75 meters underground in an extremely difficult and dangerous subterranean environment (Elliott et al. 2021). Despite bodies being smaller and thus more capable of navigating the Rising Star system than contemporary humans, the carrying of bodies and behavioral actions required to bury or place them in the various locations described would have had high energetic costs and carried substantial risk for *H. naledi*. There are no clear direct fitness benefits nor any indication of proximate functional stimuli for this suite of behaviors.

The challenges for *Homo naledi* to bury its dead in remote subterranean contexts are significant. To accomplish this *H. naledi* had to coordinate their behavior and collaborate to move the bodies to a specific location inside the cave system, excavate an area and place the body in it, and cover it, or place the remains in a specific non-floor area. The data from Berger et al. (2023a) suggest that this is a behavioral sequence that was repeated multiple times in the same location, likely across a long temporal duration. Given the complexity of the cave layout (Elliott et al. 2021), there must have been some form of explicit communication for coordination of movement and actions, and the use of fire as a light source, between the *H. naledi* undertaking the behavior. Such coordination and specific set of actions around the treatment of deceased conspecifics is more methodologically extensive, energetically costly, with higher risk of injury than any reported for other primates and non-human animals to date ([King 2013](#)). This behavior is also more complex and multifactorial than that reported for the one earlier case of hominin mortuary behavior (Sima de Los Huesos, ([Carbonell and Mosquera 2006](#))).

The subterranean environment used by *H. naledi* is not only physically challenging but is also emotionally and physiologically challenging, reflecting an engagement with difficult underground spaces not seen elsewhere in the archaeological record at this time. Dark enclosed spaces, where visual perception is curtailed, can create a state of emotional arousal profoundly affecting perceptual, cognitive and social systems ([Zuccarelli et al. 2019](#)), even with some form of illumination. While we cannot yet be certain of the exact modes, intensity and quality of the fires used by *H. naledi* in the Dinaledi subsystem, we can assume that they likely provided flickering and, at best, moderate intensity light sources. In humans, and other diurnal primates, sensory deprivation through reduced or a lack of consistent visual clues creates a heightened sensitivity to other senses as well as prompting experiences of visual disturbances, hallucinations, and disorientation ([Hodgson 2021](#)). Experiences of these types of extreme and unusual environments, though often inducing fear responses, can also facilitate powerful bonding experiences ([Steidle, Hanke, and Werth 2013](#)). This range of substantive emotional and psychological reactions may explain why experiences in deep dark caves are often associated with a sense of the transcendent in contemporary humans

(Montello and Moyes 2012) and given the broad range of sensory commonalities across diurnal anthropoids, and especially apes, they likely had comparable impacts on *H. naledi* and other Pleistocene hominins using subterranean spaces. We argue that careful and coordinated treatment of the dead on several occasions, in these environments, implies particularly strong social and emotional bonds and some shared understanding of meaning (Pettitt and Anderson 2020) in *H. naledi*.

That this high-risk, high-cost, no-overt-direct-fitness-benefit behavior was undertaken repeatedly by multiple members of the *H. naledi* community across time indicates a valued social, and likely cultural, tradition. The combination of features of the behavior (burial and associated engravings), and the context in which it was undertaken (in deep caves with the use of fire for illumination), suggests a level of cognitive/semiotic meaning-making capacity in *H. naledi* (e.g., (Kissel and Fuentes 2017; 2018)) that matches similar assessments of other populations of the genus *Homo* during the same, and later, time periods (Figure 1). This assessment of meaning-making capacity for *H. naledi* is supported by the active transformation of the ‘space’ of the Dinaledi chamber and Hill antechamber to ‘place’ (Low and Lawrence-Zúñiga 2003) through the pattern of mortuary and likely funerary behavior (e.g. (Silverman 2008) involving group collective practice, high levels of social/emotion investment, and some form of communicating the commitment to undertake the endeavor.

Recent research has produced growing evidence of multiple aspects of meaning making earlier in the archaeological record and with a wider range of hominins acknowledged to be responsible for mortuary practices (Table 1 & Fig. 1). This includes meaning making within deep caves by members of the genus *Homo* beginning after 200,000 years ago at Wonderwerk in South Africa and Bruniquel in France, both of which have evidence of fire use as well. The range of examples of meaning making extend our understanding of the cultural complexity of later Pleistocene members of the genus *Homo*. Nonetheless, the Dinaledi subsystem site is the earliest known example of the pattern of mortuary, and likely funerary, behavior that becomes increasingly common in populations of the genus *Homo* in the terminal Pleistocene (Table 1). It is critical to note that *H. naledi* is the least *Homo sapiens*-like of any hominin yet described in the Middle and Late Pleistocene of Africa.

Table 1.

Table of evidence of potential mortuary behavior in hominins.

See supplemental material for references

site	type of site	Multiple bodies	type	skeletal age of specimen(s)	Age Estimate (in ka)	species	reference
Krems-Wachtberg	open air	yes	burial	infants	31	Homo sapiens	Teschler-Nicola et al. 2020
Lake Mungo	open air	yes	burial	adults	40	Homo sapiens	Bowler et al 2003
Taramsa hills	open air	no	burial	child (8-10 yrs old)	45	Homo sapiens	Vermeersch et al. 2015
La Ferrassie	cave	yes	burial	children?	45	Neandertal	Gómez-Olivencia et al 2018
Shanidar	cave	yes	burial	adults, infants, 2-3 yr old	45	Neandertal	Pomeroy et al 2020
Mezmaiskaya Cave,	cave	no	burial	infant	45	Neandertal	Golovanova et al 1999
La Chapelle-aux-Saints 1	cave	no	burial	adult	50	Neandertal	Rendu et al 2013
Dederiyeh Cave,	cave	yes	burial	children	50	Neandertal	Akazawa et al 1999.
Régourdou Cave	cave	no	burial	adult	50	Neandertal	Maureille et al 2001
Kebara	cave	yes	burial	child and adult	55	Neandertal	Pettitt 2011
Amud	cave	yes	burial	infant and adults	60	Neandertal	Hovers et al 2000
Roc de Marsal	cave	no	Burial?	child	70	Neandertal	Maureille, B., & Knüsel, C. J. 2022
Panga ya Saidi	cave	no	burial	2.5-3 yrs old	78	Homo sapiens	Martinón-Torres et al
Qafzeh	cave	yes	burial	children and adults	100	Homo sapiens	Vandermersch & Bar-Yosef 2019
Skhul	cave	yes	burial	adults and children	110	Homo sapiens	Ronen 1976
Tabun	cave	yes	burial	adult (maybe neonate?)	120	Neandertal	Pettitt 2002

Border Cave	cave	yes	burial	adult and infants	74	Homo sapiens	d'Errico & Backwell, 2016
Sima de los huesos	pit	yes	caching/mortuary behavior	adults and children	500	Neandertal	Bischoff et al. 2003
Moula-Guercy	cave	yes	modification/mortuary behavior?		100	Neandertal	Defleur et al. 1999
Herto	open air	no	modification/mortuary behavior?	adult	160	Homo sapiens	White et al. 2003
El Sidron	cave	yes	modification/mortuary behavior?	adults and children	480	Neandertal	Rosas et al. 2006
Bodo	open air	no	modification/mortuary behavior?	adult	600	Homo sp.	White 1986
Gran dolina	cave	na	modification/mortuary behavior?	adult and children	800	Homo sp.	Fernández Jalvo et al. 1999
Sterkfontein	cave	no	modification/mortuary behavior?	adult	1635	Australopithecus	Pickering et al 2000
Krapina	cave	yes	mortuary behavior	many age ranges	130	Neandertal	Russel 1987
AL-333	open air	yes	mortuary behavior?	adults, juveniles and infants	3200	Australopithecus afarensis	Pettitt 2011

A role for emotional cognition?

This pattern of emergence of meaning making in different hominins ([Figure 1](#)), to which the *Homo naledi* burial evidence makes an important contribution, also has other implications. Most particularly it may cast light on the question of the evolution of human conscious awareness of emotions and its involvement in the processes of cognitive evolution in the genus *Homo* and in hominins in general. Humans share many building blocks of emotional cognition with other mammals, and some complex abilities with other apes. We share the same visceromotor and sensorimotor foundation for emotions with other mammals for example ([Steklis and Lane 2013](#)). Moreover, a range of common emotional responses to humans has now been documented in other primates, and particularly in apes, through measurement of heart rate and skin conductance as well as more recently pupil mimicry and infrared thermography ([Nieuwburg, Ploeger, and Kret 2021](#)). Interpersonal emotional interactions also have a common basis. Emotional contagion is apparent in monkeys and apes, and apes in particular demonstrating a level of empathy through yawning and even sympathy through active consolation ([Romero, Castellanos, and de Waal 2010](#); [Preston and de Waal 2002](#)). Diverse primate species have the cognitive ability to infer emotional meaning from expressions ([Nieuwburg, Ploeger, and Kret 2021](#)). Moreover, there is anecdotal evidence for the foundations of cognitive empathy in targeted helping within apes ([Koski and Sterck 2010](#)). However, no other mammal demonstrates the emotional conscious awareness of contemporary *Homo sapiens*. Contemporary humans possess a distinctive cognitive ability to regulate emotions by bringing feelings into “rational” thought ([Green and Spikins 2020](#)). As a result humans communicate and engage in shared intentions, and meaning-making, to a degree not seen in other animals, and demonstrate motivations to share emotions, experiences and activities with other persons ([Steklis and Lane 2013](#)).

The *H. naledi* evidence suggests that conscious emotional awareness was present in this hominin despite its small brain size. The shared and planned deposition of several bodies in the Rising Star system is evidence of a shared sets of beliefs or assumptions about meaning and action, something one would term “shared grief” in contemporary humans. The creation of engravings in the same space as the burials suggests a form of shared

memorialization, or at least shared attentions and action to alter the locations beyond the interment of the bodies. Regardless of what one terms the underlying cognitive processes associated with the burial activities of *H. naledi*, they indicate a level of conscious emotional awareness that enables and is associated with extensive shared intentionality, forward planning, and repeated cultural behavior involving bodily risk. Equally complex symbolic use of caves by Neanderthals (Jaubert et al. 2016; Baquedano et al. 2023) demonstrate a similar emotional self-awareness, and production of highly symmetrical stone tools is also potentially indicative of emotional awareness and regulation in early member of the genus *Homo* (Green and Spikins 2020). However, the fact that a small-brained hominin displays these sorts of behavior suggests that the neurological capacity enabled by a larger than 1000cc brain cannot be the only factor, or necessarily the main factor, enabling the kind of emotional cognition that is considered a central factor in human evolutionary success.

That complex emotional cognition is not unique to *Homo sapiens* should not be surprising. Social understanding of emotions is widely accepted as adaptive in an evolutionary context (Nieuwburg, Ploeger, and Kret 2021) and moreover emotional awareness is associated with better life outcomes in modern human contexts (Smith et al. 2023). That brain regions associated with socio-emotional processing are relatively enlarged in *H. naledi* despite a limited overall brain size reinforces this argument, particularly as the medial prefrontal cortex is important in reflective awareness in modern humans. A specifically *H. naledi* behavioral adaptation, reflected in the burial activities, may have depended on emotional commitments to others combined with a set of cultural beliefs/practices, a high level of emotional awareness to manage these, and in turn collaboration with extensive coordination.

What does *H. naledi* burial activity mean for understanding human evolution?

As outlined in Hawks et al. (Hawks et al. 2017) the phylogenetic relationships between *H. naledi* and other populations/taxa in the genus *Homo* remain unclear. However, *H. naledi* and some other populations of the genus *Homo* overlap temporally in the expression of meaning making behavior indicating some degree of shared socio/emotional/cognitive processes. To date, *H. naledi* is the earliest example of such actions, combining both mortuary and funerary behavior with the creation of likely symbolic engravings. Thus, it is clear that the hominins in the later Pleistocene are typified by a range of brain sizes and cranial and post-cranial morphologies and that the material record in that same time period offers increased evidence for shared meaning-making. This demonstrates that such behavior is neither “modern” nor exclusive to *Homo sapiens* (*sensu lato*). Whilst this adds further evidence to our understanding of the emergence of hominin cognition there are also wider evolutionary implications. Much like potentially independent evolution of social emotional abilities in other primates (Nieuwburg, Ploeger, and Kret 2021) the behavioral evidence for small-brained *H. naledi* may suggest that some degree of analogous rather than homologous evolution underlies social emotional complexity in humans.

The evidence for the burials by *H. naledi* creates two problems for current models of human evolution. The first is: what is the relationships between *H. naledi* mortuary/funerary behavior and that of other *Homo* taxa/populations? This can be clarified via one of three possible explanations.

1. *H. naledi* is not in the human lineage and its mortuary/funerary activities and use of fire are the result of parallel evolution/homoplasy.
2. mortuary practices, use of fire and any related meaning-making capacities are very old dating to the early Pleistocene, or even Pliocene hominins, and thus are a

homology between *H. naledi* and *H. sapiens*. or,

3. *H. sapiens* “borrowed” the mortuary and funerary behavior as a cultural practice from sympatric *H. naledi* or *vice versa*.

The second problem is how to incorporate the fact that neither absolute brain size nor encephalization quotient are necessarily correlated with the meaning-making capacities and emotional-cognition complexity associated with mortuary and funerary behavior. It is assumed that a large brain was an essential step towards a uniquely human cognition, social relationships and culture (Dunbar 2003; Muthukrishna et al. 2018). However, small-brained hominins were responsible for many key changes in human evolution. Planning and forethought in stone tool production predates the origins of *Homo* (Harmand et al. 2015) and by 1.76 million years ago multiple taxa/populations of relatively small-brained hominins were likely developing separate bifacial tool traditions (Lepre et al. 2011). It is also evident that small-brained hominins (under 800-1000cc) were those who initially expanded around, and out of Africa, crossing into eastern and Southeastern Asia (Antón, Potts, and Aiello 2014). Additionally, the use of fire emerges in excess of 1.5 million years ago conspecific only with small-brained hominins (Hlubik et al. 2019). This constellation of data is particularly significant as it demonstrates that small-brained hominins are a part of the complex hominin niche that characterizes later Pleistocene members of the genus *Homo* (Mondanaro et al. 2020).

This new evidence for complexity in *H. naledi* behavior pushes back the origins of mortuary and funerary behaviors, challenges our assumptions about the role and importance of encephalization in human evolution, and suggests that the hominin emotional, socio-cognitive niche is more significant than previously thought. This will structure how we understand and model the origins and patterns of human evolution in the future (Kissel and Fuentes 2021; Spikins 2022; Spikins et al. 2019)

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Competing interests

Authors declare that they have no competing interests.

Data and materials availability

All data, code, and materials used in the analysis are available in the SOM and in Berger et al (2023a,b)

Supplementary Materials

references for [Table 1](#) on mortuary practices

supplement for [Figure 1](#)

Supplement for [Table 1](#) on mortuary practices

Refs for [table 1](#)

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Text for Supplement **table 1** on mortuary practices

Note on **table 1**: **Table 1** was constructed by searching the literature for examples of funerary behaviors in the Paleolithic. The nature of many of these finds are contentious (1). For example, some scholars have rejected Roc-de-Marsal (2, 3) as a burial based on reevaluating the context of the site, while others include it in list of European burials (). Similarly, experts are divided as to if La Chapelle-aux-Saints should (4) or should not (5, 6) be accepted as an intentional Neandertal burial. We also list sites that are not burials but instead show possible evidence of modifying the body after death. Again, scientists disagree if these are funerary practices or, in the case of cutmarks on hominin bones, cannibalism.

Most archaeologists define burial in a way that lets them detect it archaeologically (excavate a pit, put body in pit, refill pit), but this is only one type of funerary ritual. All cultures must find ways to deal with the beginning of life and with the end of life. How to treat the dead (and how to decide when someone is really dead) is culturally specific. The symbolic practices related to death are also highly varied. Just as with foodways, deathways are mediated by how a culture sees death.

All of this is to say that the symbolic aspects around death are as important as the process of dealing with the body. We might also ask how the death related symbolic behaviors help people mourn.

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Supplementary Material

Data for Figure 1

Table used to create figure 1

site	group	type	Max.age	Min.age	Midrange	associated species	references
Bruniquel	non-functional cave use	cave	-0.176	-0.176	-0.176	Neandertals	Jaubert et al 2016
Wonderwerk	non-functional cave use	cave	n/a	n/a	-0.18	unknown archaic	Chazan and Horowitz 2009
Cueva de Ardales	non-functional cave use	cave	n/a	n/a	-0.66	Neandertals	Martí et al. 2021
Lingjing	engraved objects	engraved bone	-0.125	-0.105	-0.115	archaic Homo	Li et al 2019
Kozarnika	engraved objects	engraved bone	-1.6	-1.4	-1.5	n/a	Sirakov et al. 2010
Trinil	engraved objects	engraved shell	-0.64	-0.38	-0.51	Homo erectus	Joordens et al. 2014
Bilzingsleben	engraved objects	engraved bone	-0.37	-0.23	-0.3	Homo erectus	Mania and Mania 1988
Diepkloof Rock Shelter	engraved objects	engraved eggshell	-0.119	-0.099	-0.109	n/a	Texier et al 2013
Qafzeh	engraved objects	engraved stone	-0.118	-0.075	-0.0965	Homo sapiens	Hovers et al. 1997
Sainte-Anne	engraved objects	engraved	-0.21	-0.17	-0.19	n/a	Raynal and Seguy 1986
Oldisleben	engraved objects	engraved bone	-0.13	-0.08	-0.105	n/a	Bendarik 2006
Sima de los Huesos	mortuary behavior	mortuary	-0.6	-0.4	-0.5	Neandertals	Bischoff et al. 2007
Rising Star Cave	mortuary behavior	mortuary	-0.335	-0.241	-0.29	Homo naledi	Berger et al 2022
Skhul	mortuary behavior	mortuary	-0.13	-0.1	-0.115	Homo sapiens	Ronen 1976
Tabun	mortuary behavior	mortuary	-0.12	-0.12	-0.12	Neandertals	Pettitt, P. B. (2002)
Qafzeh	mortuary behavior	mortuary	-0.13	-0.08	-0.105	Homo sapiens	Vandermeersch and Bar-Yosef 2019
Panga ya Saidi	mortuary behavior	mortuary	-0.082	-0.074	-0.078	Homo sapiens	Martinón-Torres et al. 2021
Berekhat Ram	non-utilitarian objects	art	-0.47	-0.233	-0.35	n/a	d'Errico and Nowell 2001
Makapansgat	non-utilitarian objects	art	n/a	n/a	-3	australopithecus	Bednarik 1998
Tan-Tan	non-utilitarian objects	art	-0.5	-0.3	-0.4	n/a	Bednairk 2003
Krapina	non-utilitarian objects	?	-0.13	-0.08	-0.13	Neandertals	Radović et al 2016

Gadeb 8e	ochre use	ochre	-1.45	-0.75	-1.1	n/a	Clark and Kurashina 1979
Bizat Ruhama	ochre use	ochre	-1.96	-0.78	-1.37	n/a	Ronen et al. (1998)
Kathu Pan 1b	ochre use	ochre	-0.417	-0.417	-0.417	n/a	Watts et al. 2016
Hunsgi	ochre use	ochre	-1.5	-0.3	-0.9	n/a	Watts et al 2014
Maastricht-Belvedere C	ochre use	ochre	-0.27	-0.2	-0.235	Neandertals	Roebroeks et al 2012
GnJh-15	ochre use	ochre	-0.285	-0.285	-0.285	n/a	Denio and Mcbrearty 2003
Biddenham	ornamentation	ornamentation	-0.3	-0.3	-0.3	n/a	Bednarik 2005
Contrebandier's Cave	ornamentation	ornamentation	-0.115	-0.11	-0.1125	n/a	d'Errico et al 2009
Krapina	ornamentation	ornamentation	-0.13	-0.13	-0.13	Neandertals	Radovic et al. 2015
Border Cave	ochre use	ochre	na	na	0.2	Homo sapiens	Wadley et al. 2020
Rising Star Cave system	engraved objects	Cave wall	-0.335	-0.241	-0.29	Homo naledi	Berger et al 2023b

Table X. Data used to create Figure 1.

Refs for Figure 1

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Text for notes on table X

This table is a sampling of archaeological sites that have been suggested to show signs of what some call “symbolic behavior.” Delimitating what is and what is not symbolic has been the source of contention for many decades now (1–6). Traditionally, archaeologists have defined symbols as objects that have meanings embedded in them. Yet a symbol, by its very nature, must be interpreted within a system of meaning and discerning if something is symbolic becomes difficult without knowing the cultural context within which it has been created (7). We created this table from the published literature to demonstrate that no matter what we choose to call it, culturally-mediated behaviors predate contemporary humans. Such behaviors are found with *Homo erectus* (8), Neandertals (9) and other archaic populations (10–12).

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Reviewer #1 (Public Review):

As this experience as a reviewer has been unusual, it may be helpful to outline some relevant parameters of the task at the outset. While I was invited to review the Fuentes et al. study only, two additional papers concerning the claimed engravings and burials associated with *Homo naledi* by Berger and colleagues were also provided as components of the reviewer package. The two manuscripts presenting the archaeological evidence are accessible as preprints in bioRxiv, by Lee Berger and colleagues ('2023a, 2023b').

Unfortunately, the arguments in the Fuentes et al manuscript hinge entirely on the strength of archaeological evidence for engravings and intentional burial by *Homo naledi* (presented in the abovementioned two preprints). All inferences regarding hominin behaviour and biology of *Homo naledi*, discussed by Fuentes and colleagues, are wholly dependent on the evidence presented in the archaeology preprints being true.

Yet both of the archaeological manuscripts are unfortunately weak. In short, the claims for engravings depend on the demonstration of several elements of association that are rather standard for linking material traces found in the archaeological record with particular hominin behaviours. For the particular arguments by Berger and colleagues to be demonstrated, the traces on the rock surface need to be linked causally with hominin agency, in other words, their anthropogenic nature need to be established. The author of the engravings needs to be demonstrated as a particular hominin species (*Homo naledi* in this case), and the activity of engraving needs to have taken place ~241-335 kya. After reading the manuscript on the engravings, however, what is clear is that the scratches could as easily have been made by a modern-day farmer 50 years ago, as *Homo naledi* ~335 kya. Berger and colleagues do not present any evidence to the contrary, they simply describe their narrative as the most parsimonious scenario. A particularly curious piece of information presented as evidence is a list of individuals known to have entered the Dinaledi system in recent times (and known not to have scratched the walls, one presumes, though this is not stated).

The question of intentional burial is more complex. What we know from other widely accepted early burials is that documenting the geoarchaeological context of the hominin remains is critical to assess the likelihood of an intentional burial - this needs to be established at the outset through high quality fieldwork. Yet even the boundaries of the excavation presented in the burial manuscript appear so angled or skew relative to one another (Fig. 2a) that the individual squares look to be aligned with different XY grids, which does not instill confidence in the quality of field documentation. One can make out very little from the sediment section images - which are key to identifying intrusive features associated with burials - and the multivariate geochemical analysis of sediments is unconvincing: a scatterplot (not a biplot) should have been provided showing the geochemistry of the burial sediment samples relative to the immediately surrounding sediment characteristics. While one remains excited about the potential for a spectacular archaeological discovery within the Dinaledi cave system, unfortunately, the three manuscripts provided do not present convincing evidence to that effect.

Reviewer #2 (Public Review):

Fuentes et al. provide a detailed and thoughtful commentary on the evolutionary and behavioral implications of complex behaviors associated with a small-brained hominin, *Homo naledi*. Within the Rising Star Cave of South Africa, Berger et al. 2023a,b proposed evidence that *Homo naledi* intentionally buried their dead through complex mortuary practices and engaged in symbolic expression by engraving the cave walls in cross-hatching motifs. Two burials were identified in the Rising Star cave subsystems: Feature 1 in the Dinaledi Chamber and a feature in the Hill Antechamber. The engravings are located in the Hill Antechamber near the passageway leading into the Dinaledi chamber. The authors aimed to provide evidence for burials by (1) testing sediment samples for mineral composition from within and outside the burial feature; (2) demonstrating an interruption in the stratigraphy indicative of a "bowl-shaped" feature; (3) evaluating the anatomical coherence of the skeletal remains; (4) demonstrate matrix-supported positioning of skeletal elements; and (5) determine the compatibility of non-articulated material with decomposition and subsequent collapse. Berger et al. 2023b evaluated the engravings through high resolution photography, cross-polarization, and 3D photogrammetry. Neither article involved radiometric dating of materials. While the review by Fuentes et al. highlights important assumptions about the relationship between hominin brain size, cognition, and complex behaviors, the evidence presented by Berger et al. 2023a,b does not support the claim that *Homo naledi* engaged in burial practices or symbolic expression through wall engravings.

The major weaknesses for Berger et al. 2023a are as follows:

1. The mineral composition from sediment sampled from within Dinaledi Feature 1 is not different compared to the surrounding sediment, which is one rationale proposed by the authors that would lead to the conclusion of a burial pit. An effort to replicate the multivariate statistical analysis using the data provided in SI Table 1 by this reviewer failed, and thus, the results are not replicable.
2. The authors failed to provide clear visualizations or analysis that showed an unambiguous interruption in the stratigraphy surrounding the Dinaledi Feature 1.
3. Attempts 1 and 2 were applied solely to Dinaledi Feature 1, not the Hill Antechamber Feature.
4. Skeletal cohesion does suggest that the bodies were likely covered or protected by external environment. However, given the geological context, there is minimal opportunity for scavengers or other agents to scatter the skeletal remains within such an isolated location. Thus, this alone cannot solely support intentional burials as this line of evidence is subject to equifinality.
5. Similar to the preceding statement, evidence for matrix-supported elements was inconclusive at best. There was no mention of sedimentary rate or expectations for how quickly sediments would naturally bury the remains of whole bodies in the chamber compared with the rate of decomposition of buried remains.

The major weaknesses for Berger et al. 2023b are as follows:

1. While this is incredibly difficult to accomplish, dating rock art or other cave wall engravings is the only method to ensure that the etchings were created during the time of *Homo naledi*. Unfortunately, this was not attempted. Instead, the authors state that "This description is intended to document the discovery and provide spatial and

contextual information prior to any further analyses that may require invasive sampling." Yet, the authors assign a date to the engravings in the title of the paper. Here, the authors are generating interpretations before analyses are attempted.

2. The engravings are indeed very interesting and are likely anthropogenic in origin. However, the argument that these engravings were created by *Homo naledi* is based on the bold assumption that "No physical or cultural evidence of any other hominin population occurs within this part of the cave system, and there is no evidence that recent humans or earlier hominins ever entered any adjacent area of the cave until surveys by human cave explorers during the last 40 years." (page 6). To assume that no other individual entered the cave system from the time of *Homo naledi* until 40 years ago is an unrealistic and faulty assumption. This reviewer does not discount that the engravings could have been made by *Homo naledi*, but the evidence must be sufficient to support this statement or provide other alternatives as working hypotheses.

As a discipline, paleoanthropology aims to understand the evolutionary history of the hominin clade through fossil remains, material culture, and, most recently, ancient DNA. The methods and approaches that we as paleoanthropologists use to understand the past often bridge both the humanities and the hard sciences to create a unique understanding of our shared history. We are only limited by the conditions in which time and attrition has erased pieces of our collective story from the earth. Thus, it is our responsibility to ensure that our interpretations of the past are supported by measurable and testable means, to the best of our ability, and that hypotheses are not presented as conclusions.

Unfortunately, this is not the case for Berger et al. 2023a,b. The work presented by the authors is imprudent and incomplete and does not meet the requirements set forth by our discipline. While it is important that scholars publish their work in a dutiful timeline, it is arguably more critical for scholars to take the necessary time to ensure the integrity and resolution of the work. The consequences for rushing publications with such a significant unsubstantiated find will likely result in perilous ramifications, as it is more difficult to correct an idea than to introduce one.

Reviewer #3 (Public Review):

This paper presents the cognitive implications of claims made in two accompanying papers (Berger et al. 2023a, 2023b) about the creation of rock engravings, the intentional disposal of the dead, and fire use by *Homo naledi*. The importance of the paper, therefore, relies on the validity of the claims for the presence of socio-culturally complex and cognitively demanding behaviors that are presented in the associated papers. Given the archaeological, hominin, and taphonomic analyses in the associated papers are not adequate to enable the exceptional claims for naledi-associated complex behaviors, the inferences made in this paper are currently inadequate and incomplete.

The claimed behaviors are widely recognized as complex and even quintessential to *Homo sapiens*. The implications of their unequivocal association with such a small-brained Middle Pleistocene hominin are thus far reaching. Accordingly, the main thrust of the paper is to highlight that greater cognition and complex socio-cultural behaviors were not necessarily associated with a positively encephalized brain. This argument begs the obvious question of whether absolute brain size and/or encephalization quotient (i.e., the actual brain volume of a given species relative the expected brain size for a species of the same average body size) can measure cognitive capacity and the complexity of socio-cultural behaviors among late Middle Pleistocene hominins.

Claims for a positive correlation between absolute and/or relative brain size and cognitive ability are not common in discussions surrounding the evolution of Middle- and Late Pleistocene hominin behavior. Currently, the bulk of the evidence for early complex technological and social behaviors derives from multiple sites across South Africa and postdates the emergence of *H. sapiens* by more than 100,000 years. Such lag in the expression of complex technologies and behaviors within our species renders the brain size-implies-cognitive capacity argument moot. Instead, a rich body of research over the past several decades has focused on aspects related to socio-cultural, environmental, and even the wiring of the brain in order to understand factors underlying the expression of the capacity for greater behavioral variability. In this regard, even if the claimed evidence for complex behaviors among the small-brained naledi populations proves valid, the exploration of the specific/potential socio-cultural, neuro-structural, ecological and other factors will be more informative than the emphasis on absolute/relative brain size.

The paper presents as supporting evidence previous claims for the appearance of similar complex behaviors predating the emergence of our species, *H. sapiens*, although it does acknowledge their controversial nature. It then uses the current claims for the association of such behaviors with *H. naledi* as decisive. Given the inadequate analyses in the accompanying papers and the lack of evidence for stone tools in the naledi sites, the present claims for the expression of culturally and symbolically mediated behaviors by this small-brained hominin must be adequately established. The importance of the paper thus rests on the validity of the claimed evidence—including contextual aspects—for rock engraving, mortuary practices, and the use of fire presented in the associated two papers. The claims in both associated papers are inadequate, incomplete, and largely assumption- (rather than evidence) based. As responsible and ethical researchers, the team must return to the sites, conduct the required standard chronometric and taphonomic studies and weigh the strength of the evidence before proceeding with the current claims.

Author Response:

We would like to thank the eLife reviewers for the considerable time and effort they have invested to review these manuscripts. We have also benefited from a previous round of review of the manuscript describing the proposed burial features, which underwent two rounds of revisions in a high-impact journal over a period of approximately 8 months during 2022 and early 2023. Both sets of reviews have reflected mixed responses to the evidence we have presented, with one reviewer recommending acceptance with minor editorial revisions, two recommending acceptance with minor revisions and the fourth recommending rejection based upon similar arguments to those reflected by some of the reviewers in this current round of reviews in eLife. Ultimately the managing editor of this first journal took the decision that the review process could not be completed in a timely manner and rejected the manuscript although the submission here reflected our consideration of these reviewers suggestions.

We have chosen in this initial response to the eLife reviews to include some references to the previous anonymous reviews in order to illustrate differences of opinion and differences in revision suggestions within the review process. Our goal is to offer maximal insight into our decision-making process and to acknowledge the considerable time and effort put into the assessment of these manuscripts by reviewers (for eLife and in the case of the earlier review process). We hope that this approach will assist the readers, and reviewers, of our manuscripts in understanding why we are proceeding with certain decisions during the revision process.

This is a new process for us and the reviewers, and one way in which it significantly differs from more traditional review is that both the reviews and our reply will be public well in advance of our revisions to the manuscript. Indeed, considering the scope of the reviews, some of those revisions may take considerable time, although many can be accomplished fairly easily. Thus, we are not in a position to say that we have solved every issue raised by the reviewers. Instead, we will examine what appear to be the key critical issues raised regarding the data and the analyses and how we propose to address these as we revise the papers. We will also address several philosophical and ethical issues raised by the reviews and our proposal for dealing with these. More specific editorial and citational recommendations will be dealt with on a case-by-case basis, and we do not address these point-by-point in this reply. Please note, this response to the reviewers is not the revision of the manuscript and is only the initial opinion of the corresponding authors with some guidance from the larger group of authors of all three papers. Our final submitted revision will reflect the input of all authors included on those submissions.

We took the decision to submit three separate papers consciously. The two different categories of evidence, burials and engravings, involve different kinds of analysis and different (although overlapping) teams of researchers, and we recognized that each deserved their own presentation and assessment. Meanwhile, together they inform the context of *H. naledi* in a way that requires some synthetic discussion, in which both kinds of evidence are relevant, leading to a third paper. But the mutual relevance of these different kinds of evidence and their review by a common set of reviewers naturally raises cross-cutting issues, and the reviewers have cross-referenced the three articles. This has sometimes led to suggestions about one manuscript based on the contents of another. Considering the situation, we accepted the recommendation that it would be clearer to consider all three articles in a single reply. Thus, while each of the three papers will proceed separately during the revision process, it will be necessary to highlight across all three papers occasionally in our responses.

Scientific Issues:

In reading the reviews, we feel there are 9 critical points/assertions raised by one or more of the reviewers that present a problem for, or challenge to, our hypothesis that the observed evidence (bone accumulations and engravings) described in the Dinaledi subsystem are of intentional naledigenic origin. These are:

1. The evidence presented does not demonstrate a clear interruption of the floor sediments, thus failing to demonstrate excavated holes.
2. The sediments infilling the holes where the skeletal remains are found have not been demonstrated to originate from the disruption of the floor sediments and thus could be part of a natural geological process (e.g. water movement, slumping) or carnivore accumulations.
3. Previous geological interpretations by our research group have given alternative geological explanations for formation of the bony accumulations that contradict the present evidence presented here and result in alternative origins hypotheses.
4. Burial cannot be effectively assessed without complete excavation of the features and site.
5. The skeletal remains as presented do not conform clearly to typical body arrangement/positions associated with human (*Homo sapiens*) burials.

6. There is no evidence of grave goods or lithic scatters that are typically associated with human burials.
7. Humans may have been involved with the creation of either the *Homo naledi* bone accumulations, the engravings, or both.
8. Without a date of the engravings, the null hypothesis should be the engravings were created by *Homo sapiens*.
9. The null hypothesis for explanation of the skeletal remains in this situation should be “natural accumulation”.

Our analysis of the Dinaledi Feature 1 leads us to accept that the laminated orange-red mudstone (LORM) sedimentary layer is interrupted, indicating a non-natural intervention, and that the hole created by the interruption was then filled by both a fleshed body (and perhaps parts of other bodies) which were then covered by sediment that originated from the hole that was dug. We recognize that the four eLife reviewers are not convinced that our presentation is sufficient to establish this. Interestingly, this was not the universal opinion of earlier reviewers of the initial manuscript several of whom felt we had adequately supported this hypothesis. The lack of clarity in this current version of the burial manuscript is our responsibility. In the upcoming revision of this paper to be submitted, we will take the reviewers’ critiques to heart and add additional figures that illustrate better the disruption of the LORM and clarify the sedimentological data showing the material covering the skeletal remains in the hole are the disrupted sediments excavated from the same hole. We are proposing to isolate this most critical evidence for burial into a separate section in the revised submission based on the reviewers’ comments. The fact that the LORM layer is disrupted, a fleshed body was placed in the hole created by this disruption, and the body (and perhaps parts of other bodies) was/were then covered by the same sediments from the hole is the central feature of our hypothesis that the bone accumulations observed reflect a burial and not a natural process.

The possibility of fluvial transport or involvement in the subsystem is a topic that we have addressed extensively in past work, and it is clear from these reviews that we must enhance our current manuscript to discuss this issue at greater length. Our previous work (Dirks et al. 2015; Dirks et al. 2017) emphasized that fluvial transport of whole bodies into the subsystem was precluded by several lines of sedimentological evidence. We excavated a rich accumulation of skeletal remains, including articulated limbs and other elements in subvertical orientations inconsistent with slow sedimentary infill, which were difficult to explain without positing either a large and dense pile of bodies and/or sediment movement. We encountered fractured chunks of laminated orange-red mudstone (LORM) in random orientations within our excavation area, within and among skeletal remains, which directly refuted that the remains were inundated with water at the time of burial, and this limited the possibility of fluvial transport. Water flow sufficient to displace bodies or complete skeletal evidence would also transport large and coarse sediment, which is absent from the subsystem, and would sort the commingled skeletal material that we found by size, which we do not observe. But our excavation only covered less than a square meter at very limited depth, and this was the limit to our knowledge of subsurface sediment. We thus were left with uncertainty that led us to suggest the possibility of sediment slumping or movement into subsurface drains, although these were not observed near our excavation. Our current work expands our knowledge of the subsurface and presents an alternative explanation for the disposition of skeletal remains from our earlier excavation. But we acknowledge that this new explanation is vulnerable to our own previous published proposals, and we must do a better job of explaining how the new information addresses our previous suggestions. By not clearly creating a section where we explained how these previous hypotheses were now nullified by

new evidence, we clearly confused the reviewers with our own previous work. We will revise the manuscript by enhancing the review of the significant geological evidence demonstrating that there is no significant fluvial action in the system and making it clear how the burial hypothesis provides a clearer explanation for the situation of skeletal remains from our previous excavation work.

One of the central issues raised by reviewers has been a perceived need to excavate these features completely, totally exhuming all skeletal remains from them. Reviewers have written that it is necessary to identify every skeletal element that is present and account for any missing elements. On this point, we have both ethical and scientific differences from these reviewers. We express our ethical concerns first. Many of the best-preserved possible burials ever discovered by archaeologists were subjected to total excavation and exhumation. Cases like La Chapelle-aux-Saints, La Ferrassie, and Skhūl were fully excavated at a time when data recording and excavation methods did not include the range of spatial and geomorphological approaches that later became routine. The judgment of early investigators that these situations were intentional burials was challenged by later workers, and the kind of information that might enable better tests had been irrevocably lost (Gargett 1999; Dibble et al. 2015; Rendu et al. 2014).

Later, improved excavation standards have not sufficed to remove uncertainty or debate about possible burials. For example, it was long presumed that well-preserved remains of young children were by themselves diagnostic of intentional burial, such as those from Dederiyeh, Border Cave, or Roc de Marsal. Such cases were also fully excavated, with adequate documentation of the positioning of skeletal remains and their surrounding stratigraphic situation, but such cases were later challenged on several bases and the complete exhumation of material has confused or precluded testing of new hypotheses (e.g. Gargett 1999). The case of Roc de Marsal is one in which data from the initial excavation combined with data from the initial excavation combined with re-excavation and geoarchaeological analysis led to a naturalistic interpretation of the skeletal material (Sandgathe et al. 2011; Goldberg et al. 2017). But even in this case, the researchers erred in their interpretation of the skeleton's situation due to a lack of identification of parts of the infant's skeleton (Gómez-Olivencia and García-Martínez 2019). That is to say, it is not only the burial hypothesis but other hypotheses that suffer from complete excavation. Researchers concerned with preserving all possible information have sometimes taken extraordinary measures to remove and study possible burials at high-resolution in the laboratory. Such was the case of the Shanidar IV burial removed from the site and transported in plaster jacket by Solecki, which led to the disruption and loss of internal stratigraphic information (Pomeroy et al. 2020). Arguably, the current state of the art is full excavation with partial preparation, such as that undertaken at Panga ya Saidi (Martínón-Torres et al. 2021). But again, any future attempt to reinterpret or test the hypothesis of burial must rely on the adequacy of documentation as the original context has been removed.

In our decision to leave material in place as much as possible, we are expanding upon standard practice to leave witness sections and unexcavated areas for future research. The situation is novel, representing possible burials by a nonhuman species, and that makes it doubly important in our opinion to be conservative in not fully exhuming the skeletal material from its context. We anticipate that many other researchers, including future investigators, will suggest additional methods to further test the hypothesis of burial, something that would be impossible if we had excavated the features in their entirety prior to publishing a description of our work. We believe strongly that our ethical responsibility is to publish the work and the most likely interpretation while leaving as much evidence in place as possible to enable further testing and replication. We welcome the suggestions of additional methods/analyses to test the *H. naledi* burial hypothesis.

This being said, we also observe that total exhumation would not resolve the concerns raised by the reviewers. The recommendation of total exhumation is in pursuit of a full account of all skeletal material present and its preservation and spatial situation, in order to demonstrate that they conform to body positions comparable to human burials. As has been highlighted in forensic casework, the excavation of an inhumation feature does not necessarily provide an accurate spatial or anatomical manifest of the stratigraphical relationships between the body, encapsulating matrix, and any cut present due to preservational, taphonomic and operational factors (Dirkmaat and Cabo, 2016; Hunter, 2014). In particular, in cases where skeletal elements are highly fragmented, friable, or degraded (such as through bioerosion) then complete excavation—even under controlled laboratory conditions—may destroy bone and severely limit skeletal identification (Henderson, 1997; Hochrein, 2002; Owsley and Compton, 1997), particularly in elements where the ratio of trabecular to cortical bone is high (Darwent and Lyman, 2002; Lyman, 1994). As such, non-invasive methods of 3D and 4D modelling (preservation *in situ*) are often considered preferable to complete necropsy or excavation (preservation by record) where appropriate (Bolliger and Thali, 2009; Dell'Unto and Landeschi, 2022; Randolph-Quinney *et al.*, 2018; Silver, 2016).

The test of burial is not primarily positional, but taphonomic and geological. The position and number of bones can elaborate on process-driven questions of decay and destruction in the burial environment, or post-mortem modification, but are not singularly indicative of whether the remains were intentionally buried – the post-mortem narrative of *all* the processes affecting the cadaveric island is required (Knüsel and Robb, 2016). In previous cases, researchers have disputed or accepted the hypothesis of intentional hominin burial based upon assumptions about how modern humans or Neandertals would have positioned bodies, with the idea that some positions reflect ritual intent while others do not. But applying such assumptions is unjustifiable, particularly for a species like *H. naledi*, whose culture may have differed fundamentally from our own. Our work acknowledges that the present evidence does not enable a full reconstruction of the burial positions, but it does show that fleshed remains were encased in sediment prior to decomposition of soft tissue, and that subsequent spatial changes can be most parsimoniously explained by natural decomposition within sedimentary matrix contained within a burial feature (after Green, 2022; Mickleburgh and Wescott, 2018; Mickleburgh *et al.*, 2022). If the argument is that extraordinary claims require extraordinary evidence, we feel that the evidence documents excavation and interment (and will do so more clearly in the revision) and the fact of the remains do not match a “typical” human burial in body positioning is not in itself evidence that these are not *H. naledi* burials.

We feel that the reviewers (in keeping with many palaeoanthropologists) have a clear idea of what they “think” a burial should look like in an idealised sense, but this platonic ideal of burial form is not matched by the extensive literature in archaeoethnology, funerary archaeology and forensic science which indicates enormous variability in the activity, morphology and post-mortem system experienced by the human body in cases of interment and body disposal (e.g. Aspöck, 2008; Boulestin and Duda, 2005 and 2006; Connelly *et al.*, 2005; Channing and Randolph-Quinney, 2006; Cherryson, 2008; Donnelly *et al.*, 1995; Finley, 2000; Hunter, 2014; Parker Pearson, 1999; Randolph-Quinney, 2013). Decades of experience in the identification, recovery and interpretation of clandestine, deviant, and non-formal burials indicates the platonic ideal is rare, and in many contexts, the exception (Cherryson, 2008; Parker Pearson, 1999). This variability is particularly relevant to morphological traits in burial context, such as the informal nature of the grave cut in plan and section, shallow burial depth, and initial disposition of body (placement) during the early post-mortem period. These might run counter to the expectations of reviewers or others referencing the fossil hominin record, but are well accepted within the communities of researchers investigating Holocene archaeological sites and forensic contexts.

It is encouraging to see reviewers beginning to incorporate the extensive (often experimentally derived) literature from archaeothanatology and forensic taphonomy in their deliberations, and we will be taking these comments on board going forward. In particular, we acknowledge reviewers' comments and the need to construct a more detailed post-mortem narrative, accounting for joint disarticulation (labile versus persistent joints etc), displacement, and final disposition of elements within the burial space. As such we will incorporate the hierarchy of decomposition (rank order disarticulation), associations between regions of anatomical association, areas of disassociation, and the voids produced during decomposition (after Mickleburgh and Wescott, 2018; Mickleburgh et al., 2022) into our narrative. In doing so we acknowledge the tensions between the inductive archaeothanatological narrative-driven approach (e.g. Duday, 2005 & 2009) versus robust decomposition data derived from human forensic taphonomic experimentation recently articulated by Schotsmans and colleagues (2022) - noting that we will highlight comparative data based on forensic experimental casework and actualistic modelling over inductive intuitive approaches which come with significant evidential shortcomings (Bristow *et al.* 2011).

Finally, from a taphonomic perspective it is worth pointing out to reviewers that we have already addressed the issue of lack of taphonomic evidence for carnivore involvement in the formation of the Dinaledi assemblage (Dirks, *et al.*, 2016). Absence of any carnivore-induced bone surface modifications, patterns of skeletal part representation, and a total absence of any carnivore remains found within the Dinaledi chamber (following Kuhn and colleagues, 2010) lead us to reject carnivores as possible vectors of body accumulation within the Dinaledi Chamber and Hill Antechamber.

Reviewers suggest that without a date derived from geochronological methods, the engravings cannot be associated with *H. naledi*, and that it is possible (or probable) that the engravings were done in the recent past by *H. sapiens*. This suggestion neglects the context of the site. We have previously documented the structure and extremely limited accessibility of the Dinaledi subsystem. This subsystem was not recorded on maps of the documented Rising Star Cave system prior to our work and its discovery by our teams. Furthermore, there is no evidence of prehistoric human activity in the areas of the cave related to possible subterranean entrances. There is no evidence that humans in the past typically ventured into such extreme spaces like those of Rising Star. It is clear from the presence of the remains of many individuals that *H. naledi* ventured into these spaces again and again. It is likely that *H. naledi* moved through these spaces more easily than humans do based on their physique. We show that the engravings overlay each other suggesting multiple engraving events. These engravings took time and effort and the only evidence for use of the Dinaledi subsystem by any hominin is by *H. naledi*. The context leads to the null hypothesis that *H. naledi* made the marks. In our revision, we will elaborate on this argument to clarify the evidence for our stance on this hypothesis. Several reviewers took issue with the title of the engraving paper as we did not insert a qualifier in front of the suggested date range for the engravings. We deliberately left out qualifying language so that the title took the form of a testable hypothesis rather than a weak assertion. Should future work find the engravings were not produced within this time range, then we will restate this hypothesis.

Finally, with regards to the engravings we have chosen to report them because they exist. Not reporting the presence of engraved marks on the walls of a cave above hypothesized burials would be tantamount to leaving relevant evidence out of the description of an archaeological context. We recognize and state in our manuscript that these markings require substantial further study, including attempts at geochronological dating. But the current evidence is clearly relevant to the archaeological context of the subsystem. We take a similar stance with reporting the presence of the tool shaped artefact near the hand of the *H. naledi* skeleton in the Hill Antechamber. It is evident that this object requires further study, as we stated in our manuscript, but again omitting it from our study would be leaving out relevant evidence.

Some have suggested that the null hypothesis should be that all of these observed circumstances are of natural origin. Our team took this approach in our early investigation of the Dinaledi subsystem (Dirks et al. 2015). We adopted the null hypothesis that the geological processes involved in the accumulation of *H. naledi* skeletal remains were “natural” (e.g., non-naledigenic involvement), and we were able to reject many alternative explanations for the assemblage, including carnivore accumulation, “death trap” accumulation, and fluvial transport of bodies or bones (Dirks et al. 2015). This led us to the hypothesis that *H. naledi* were involved in bringing the bodies into the spaces where they were found. But we did not hypothesize their involvement in the formation of the deposit itself beyond bringing the bodies to the location.

This approach seems conservative. It followed the traditional view that small-brained hominins do not engage in cultural practices. But we recognize in hindsight that this null hypothesis approach did harm to our analyses. It impeded us from recognizing within our initial excavations of the puzzle box area and other excavations between 2014 – 2017 that we might be encountering remains that were intrusive in the sedimentary floor of the chamber. If we had approached the accumulation of a large number of hominins from the perspective of the null hypothesis being that the situation was likely cultural, we perhaps would have collected evidence in a slightly different manner. We certainly note that if the Dinaledi system had been full of the remains of modern humans, there would have been little doubt that the null hypothesis would have been that this was a cultural space and not a “natural space”. We therefore respectfully disagree with the reviewers who continue to support the idea that we should approach hominin excavations with the null hypothesis that they will be natural (specifically non-cultural) in origins. If excavations continue with this mindset we believe that potential cultural evidence is almost certain to be lost.

There has been a gradient across paleoanthropological excavations, archaeological work, and forensic investigation, with increasing precision of context. The reality is that the recording precision and frame of approach is typically different in most paleontological excavations than in those related to contemporary human remains. If anything comes from the present discussion of whether the Dinaledi system is a burial site for *H. naledi* or not, we hope that by taking seriously the possibility of deep cultural dynamics of hominins, we will encourage other teams to meet the highest standards of excavation in order to preserve potential cultural evidence. Given *H. naledi*'s cranial capacity we suggest that even very early hominin skeletal assemblages should be re-examined, if there is sufficient evidence or records available. These would include examples such as the A.L. 333 *Au. afarensis* site (the so called First Family site in Hadar Ethiopia), the Dikika infant skeleton, WT 15000 (Turkana Boy) and even A.L. 288 (Lucy) as such unusual taphonomic situations where skeletons are preserved cannot be simply explained away as “natural” in origin, based solely on the cranial capacity and assumed lack of cognitive and cultural complexity of the hominins as emphasized by us in Fuentes et al. (2023). We are not the first to observe that some very early hominin situations may represent early mortuary activity (Pettitt 2013), but we would advocate a step further. We suggest it may be damaging to take “natural accumulation” as the standard null hypothesis for hominin paleoanthropology, and that it is more conservative in practice to engage remains with the null hypothesis of possible cultural formation.

We are deeply grateful for the time and effort all of the 8 reviewers (across three reviews) have taken with this work. We also acknowledge the anonymous reviewers from previous submissions who's opinions and comments will have made the final iterations of these manuscripts better for their efforts. As this process is rather public and includes commentary outside of the *eLife* forum, we ask that the efforts of all 37 authors and 8 reviewers involved be respected and that the discourse remain professional in all venues as we study this fascinating and quite complex occurrence. We appreciate also the efforts of members of the public who have engaged with this relatively new process where preprints are posted prior to

the reviews allowing comments and interactions from colleagues and the public who are normally not part of the internal peer review process. We believe these interactions will make for better final papers. We feel we have met the standards of demonstrating burials in *H. naledi* and that the engravings are most likely associated with *H. naledi*. However, given the reviews we see many areas where our clarity and context, and analyses, were less strong than they can be. With the clarifications and additions taken on board through these review processes the final papers will be stronger and clearer. We recognize that this is an ongoing process of scientific investigation and further work will allow continued, and possibly better, evaluation of these hypothesis and others.

Lee R Berger, Agustín Fuentes, John Hawks, Tebogo Makhubela

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